Classical theories of information science

信息度量
Information Measures

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Outline

1. Independence and Markov Chain
2. Shannon's Information Measures
   - Entropy, Conditional Entropy
   - Mutual Information, Conditional Mutual Information
3. Chain Rules
4. Information Divergence
5. The Basic Inequalities

4. Information Divergence

Informational Divergence

Definition 2.28. *The informational divergence between two probability distributions p and q on a common alphabet $\mathcal{X}$ is defined as*

$$D(p\|q) = \sum_x p(x) \log \frac{p(x)}{q(x)} = E_p \log \frac{p(X)}{q(X)}, \tag{2.104}$$

where E_p denotes expectation with respect to p.

- Convention:
  1. summation over $\mathcal{S}_p$
  2. $c \log \frac{c}{0} = \infty$ for $c > 0$ — if $D(p\|q) < \infty$, then $\mathcal{S}_p \subset \mathcal{S}_q$.
- $D(p\|q)$ measures the "distance" between p and q.
- $D(p\|q)$ is not symmetrical in p and q, so $D(\cdot\|\cdot)$ is not a true metric.
- $D(\cdot\|\cdot)$ does not satisfy the triangular inequality.
- Also called *relative entropy* or the *Kullback-Leibler distance*.

Fundamental Inequality

Lemma 2.29 (Fundamental Inequality). *For any $a > 0$,*

$$\ln a \leq a - 1 \tag{2.105}$$

with equality if and only if $a = 1$.

Proof. Let $f(a) = \ln a - a + 1$. Then $f'(a) = 1/a - 1$ and $f''(a) = -1/a^2$. Since $f(1) = 0$, $f'(1) = 0$, and $f''(1) = -1 < 0$, we see that $f(a)$ attains its maximum value 0 when $a = 1$. This proves (2.105). It is also clear that equality holds in (2.105) if and only if $a = 1$. Figure 2.3 is an illustration of the fundamental inequality. □

Fundamental Inequality

Lemma 2.29 (Fundamental Inequality). *For any $a > 0$,*

$$\ln a \leq a - 1 \tag{2.105}$$

with equality if and only if $a = 1$.

Corollary 2.30. *For any $a > 0$,*

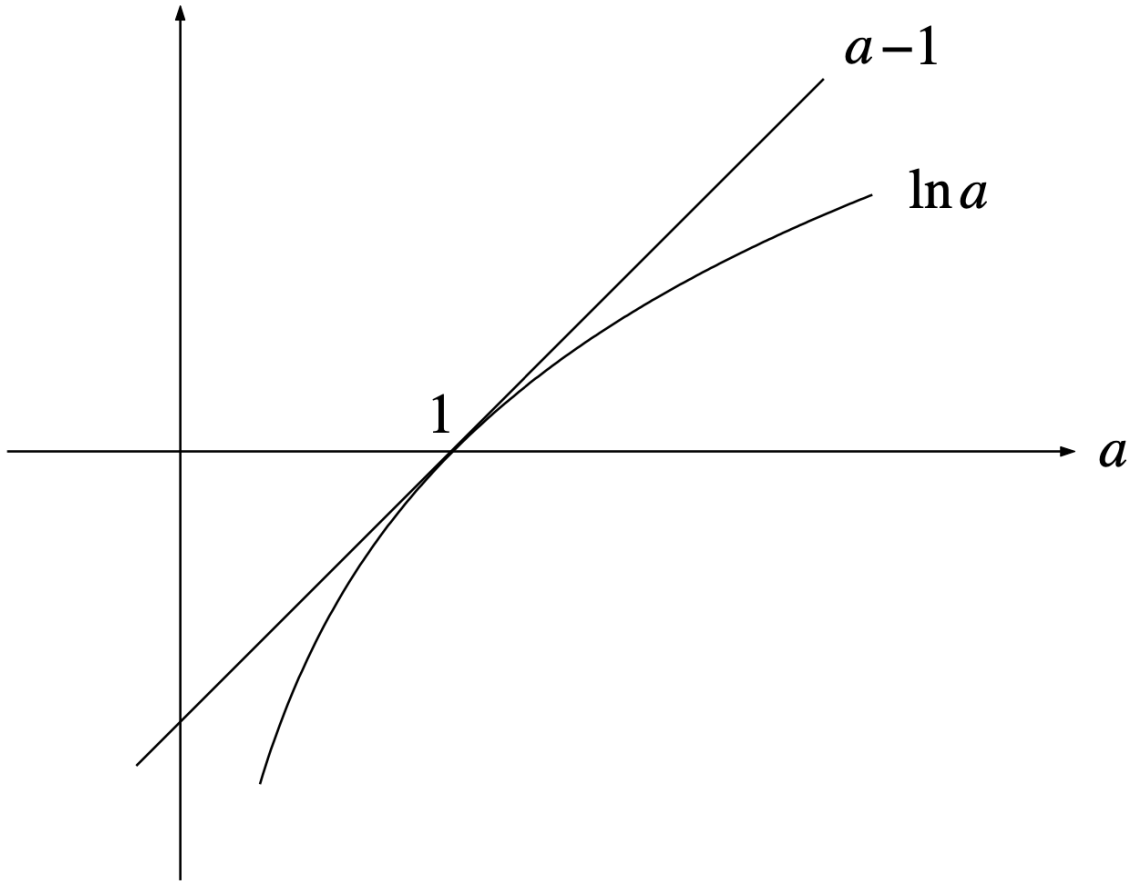
$$\ln a \geq 1 - \frac{1}{a} \tag{2.106}$$

with equality if and only if $a = 1$.

Proof. This can be proved by replacing a by $1/a$ in (2.105). □

Visualization of Fundamental Inequality

$$\ln a \leq a - 1$$



Divergence Inequality

Theorem 2.31 (Divergence Inequality). *For any two probability distributions p and q on a common alphabet $\mathcal{X}$,*

$$D(p\|q) \geq 0 \tag{2.107}$$

with equality if and only if $p = q$.

Proof. If $q(x) = 0$ for some $x \in \mathcal{S}_p$, then $D(p\|q) = \infty$ and the theorem is trivially true.

Therefore, we assume that $q(x) > 0$ for all $x \in \mathcal{S}_p$.

Consider

$$D(p\|q) = (\log e) \sum_{x \in \mathcal{S}_p} p(x) \ln \frac{p(x)}{q(x)} \tag{2.108}$$

$$\geq (\log e) \sum_{x \in \mathcal{S}_p} p(x) \left(1 - \frac{q(x)}{p(x)}\right) \tag{2.109}$$

$\ln a \geq 1 - \frac{1}{a}$

$$= (\log e) \left[\sum_{x \in \mathcal{S}_p} p(x) - \sum_{x \in \mathcal{S}_p} q(x)\right] \tag{2.110}$$

$$\geq 0, \tag{2.111}$$

where (2.109) results from an application of (2.106), and (2.111) follows from $\sum_{x \in \mathcal{S}_p} q(x) \leq 1 = \sum_{x \in \mathcal{S}_p} p(x)$. This proves (2.107).

Theorem 2.31 (Divergence Inequality). *For any two probability distributions p and q on a common alphabet $\mathcal{X}$,*

$$D(p\|q) \geq 0 \tag{2.107}$$

with equality if and only if $p = q$.

Proof (continue):

For equality to hold in (2.107), equality must hold in (2.109) for all $x \in \mathcal{S}_p$ and also in (2.111). For the former, we see from Lemma 2.29 that this is the case if and only if

$$p(x) = q(x) \quad \text{for all } x \in \mathcal{S}_p, \tag{2.113}$$

which implies

$$\sum_{x \in \mathcal{S}_p} q(x) = \sum_{x \in \mathcal{S}_p} p(x) = 1, \tag{2.114}$$

i.e., (2.111) holds with equality. Thus (2.113) is a necessary and sufficient condition for equality to hold in (2.107).

It is immediate that $p = q$ implies (2.113), so it remains to prove the converse. Since $\sum_x q(x) = 1$ and $q(x) \geq 0$ for all x, $p(x) = q(x)$ for all $x \in \mathcal{S}_p$ implies $q(x) = 0$ for all $x \notin \mathcal{S}_p$, and therefore $p = q$. The theorem is proved. □

Divergence Inequality

Theorem 2.32 (Log-Sum Inequality). *For positive numbers $a_1, a_2, \cdots$ and nonnegative numbers $b_1, b_2, \cdots$ such that $\sum_i a_i < \infty$ and $0 < \sum_i b_i < \infty$,*

$$\sum_i a_i \log \frac{a_i}{b_i} \geq \left(\sum_i a_i \right) \log \frac{\sum_i a_i}{\sum_i b_i} \tag{2.115}$$

with the convention that $\log \frac{a_i}{0} = \infty$. Moreover, equality holds if and only if $\frac{a_i}{b_i} = constant$ for all i.

Example: $a_1 \log \frac{a_1}{b_1} + a_2 \log \frac{a_2}{b_2} \geq (a_1 + a_2) \log \frac{a_1 + a_2}{b_1 + b_2}.$

Proof

Theorem 2.32 (Log-Sum Inequality). *For positive numbers $a_1, a_2, \cdots$ and nonnegative numbers $b_1, b_2, \cdots$ such that $\sum_i a_i < \infty$ and $0 < \sum_i b_i < \infty$,*

$$\sum_i a_i \log \frac{a_i}{b_i} \geq \left(\sum_i a_i\right) \log \frac{\sum_i a_i}{\sum_i b_i} \tag{2.115}$$

with the convention that $\log \frac{a_i}{0} = \infty$. Moreover, equality holds if and only if $\frac{a_i}{b_i} =$ constant for all i.

Proof of Theorem 2.32. Let $a'_i = a_i / \sum_j a_j$ and $b'_i = b_i / \sum_j b_j$. Then $\{a'_i\}$ and $\{b'_i\}$ are probability distributions. Using the divergence inequality, we have

$$0 \leq \sum_i a'_i \log \frac{a'_i}{b'_i} \tag{2.117}$$

$$= \sum_i \frac{a_i}{\sum_j a_j} \log \frac{a_i / \sum_j a_j}{b_i / \sum_j b_j} \tag{2.118}$$

$$= \frac{1}{\sum_j a_j} \left[\sum_i a_i \log \frac{a_i}{b_i} - \left(\sum_i a_i\right) \log \frac{\sum_j a_j}{\sum_j b_j}\right], \tag{2.119}$$

which implies (2.115). Equality holds if and only if $a'_i = b'_i$ for all i, or $\frac{a_i}{b_i} =$ *constant* for all i. The theorem is proved. □

Divergence Inequality vs Log-Sum Inequality

- The divergence inequality implies the log-sum inequality.
- The log-sum inequality also implies the divergence inequality.
- The two inequalities are equivalent.

5. The Basic Inequalities

The basic inequalities

Theorem 2.34. *For random variables X, Y, and Z,*

$$I(X;Y|Z) \geq 0, \tag{2.121}$$

with equality if and only if X and Y are independent when conditioning on Z.

Corollary All Shannon's information measures are nonnegative, because they are all special cases of conditional mutual information.

Proposition 2.35 $H(X) = 0$ if and only if X is deterministic.

Proposition 2.36 $H(Y|X) = 0$ if and only if Y is a function of X.

Proposition 2.37 $I(X;Y) = 0$ if and only if X and Y are independent.

Theorem 2.34. *For random variables X, Y, and Z,*

$$I(X;Y|Z) \geq 0, \tag{2.121}$$

with equality if and only if X and Y are independent when conditioning on Z.

Proof. Observe that

$$I(X;Y|Z)$$

$$= \sum_{x,y,z} p(x,y,z) \log \frac{p(x,y|z)}{p(x|z)p(y|z)} \tag{2.122}$$

$$= \sum_{z} p(z) \sum_{x,y} p(x,y|z) \log \frac{p(x,y|z)}{p(x|z)p(y|z)} \tag{2.123}$$

$$= \sum_{z} p(z) D(p_{XY|z} \| p_{X|z} p_{Y|z}), \tag{2.124}$$

where we have used $p_{XY|z}$ to denote $\{p(x,y|z), (x,y) \in \mathcal{X} \times \mathcal{Y}\}$, etc. Since for a fixed z, both $p_{XY|z}$ and $p_{X|z}p_{Y|z}$ are joint probability distributions on $\mathcal{X} \times \mathcal{Y}$, we have

$$D(p_{XY|z} \| p_{X|z} p_{Y|z}) \geq 0. \tag{2.125}$$

Therefore, we conclude that $I(X;Y|Z) \geq 0$. Finally, we see from Theorem 2.31 that $I(X;Y|Z) = 0$ if and only if for all $z \in \mathcal{S}_z$,

$$p(x,y|z) = p(x|z)p(y|z), \tag{2.126}$$

or

$$p(x,y,z) = p(x,z)p(y|z) \tag{2.127}$$

for all x and y. Therefore, X and Y are independent conditioning on Z. The proof is accomplished. □

Conditioning does not increase entropy

Theorem 2.38 (Conditioning Does Not Increase Entropy)

$$H(Y|X) \leq H(Y)$$

with equality if and only if X and Y are independent.

- Similarly, $H(Y|X,Z) \leq H(Y|Z)$.
- Warning: $I(X;Y|Z) \leq I(X;Y)$ does not hold in general.

Proof. This can be proved by considering

$$H(Y|X) = H(Y) - I(X;Y) \leq H(Y), \tag{2.132}$$

where the inequality follows because $I(X;Y)$ is always nonnegative. The inequality is tight if and only if $I(X;Y) = 0$, which is equivalent by Proposition 2.37 to X and Y being independent. □

Independence Bound for Entropy

Theorem 2.39 (Independence Bound for Entropy).

$$H(X_1, X_2, \cdots, X_n) \leq \sum_{i=1}^{n} H(X_i) \tag{2.134}$$

with equality if and only if X_i, $i = 1, 2, \cdots, n$ are mutually independent.

Proof. By the chain rule for entropy,

$$H(X_1, X_2, \cdots, X_n) = \sum_{i=1}^{n} H(X_i|X_1, \cdots, X_{i-1}) \tag{2.135}$$

$$\leq \sum_{i=1}^{n} H(X_i), \tag{2.136}$$

where the inequality follows because we have proved in the last theorem that conditioning does not increase entropy.

The inequality is tight if and only if it is tight for each i, i.e.,

$$H(X_i|X_1, \cdots, X_{i-1}) = H(X_i) \tag{2.137}$$

for $1 \leq i \leq n$.

From the last theorem, this is equivalent to X_i being independent of $X_1, X_2, \cdots, X_{i-1}$ for each i. Then

$$p(x_1, x_2, \cdots, x_n)$$

$$= p(x_1, x_2, \cdots, x_{n-1})p(x_n) \tag{2.138}$$

$$= p(p(x_1, x_2, \cdots, x_{n-2})p(x_{n-1})p(x_n) \tag{2.139}$$

$$\vdots$$

$$= p(x_1)p(x_2)\cdots p(x_n) \tag{2.140}$$

for all $x_1, x_2, \cdots, x_n$, i.e., $X_1, X_2, \cdots, X_n$ are mutually independent.

Independence Bound for Entropy

Alternatively, we can prove the theorem by considering

$$\sum_{i=1}^{n} H(X_i) - H(X_1, X_2, \cdots, X_n)$$

$$= -\sum_{i=1}^{n} E \log p(X_i) + E \log p(X_1, X_2, \cdots, X_n) \tag{2.141}$$

$$= -E \log[p(X_1)p(X_2)\cdots p(X_n)] + E \log p(X_1, X_2, \cdots, X_n) \tag{2.142}$$

$$= E \log \frac{p(X_1, X_2, \cdots, X_n)}{p(X_1)p(X_2)\cdots p(X_n)} \tag{2.143}$$

$$= D(p_{X_1 X_2 \cdots X_n} \| p_{X_1} p_{X_2} \cdots p_{X_n}) \tag{2.144}$$

$$\geq 0, \tag{2.145}$$

where equality holds if and only if

$$p(x_1, x_2, \cdots, x_n) = p(x_1)p(x_2)\cdots p(x_n) \tag{2.146}$$

for all $x_1, x_2, \cdots, x_n$, i.e., $X_1, X_2, \cdots, X_n$ are mutually independent. □

Useful Inequalities

Proposition 2.26 (Chain Rule for Mutual Information).

$$I(X_1, X_2, \cdots, X_n; Y) = \sum_{i=1}^{n} I(X_i; Y | X_1, \cdots, X_{i-1}). \tag{2.99}$$

Theorem 2.40

$$I(X; Y, Z) \geq I(X; Y),$$

with equality if and only if $X \to Y \to Z$ forms a Markov chain.

Proof. By the chain rule for mutual information, we have

$$I(X; Y, Z) = I(X; Y) + I(X; Z|Y) \geq I(X; Y). \tag{2.148}$$

The above inequality is tight if and only if $I(X; Z|Y) = 0$, or $X \to Y \to Z$ forms a Markov chain. The theorem is proved. □

Useful Information Inequalities

Lemma 2.41 If $X \to Y \to Z$ forms a Markov chain, then

$$I(X;Z) \leq I(X;Y)$$

and

$$I(X;Z) \leq I(Y;Z).$$

Corollary

- If $X \to Y \to Z$, then $H(X|Z) \geq H(X|Y)$.
- Suppose Y is an observation of X. Then further processing of Y can only increase the uncertainty about X on the average.

Proof of Lemma 2.41

Theorem 2.34. *For random variables X, Y, and Z,*

$$I(X;Y|Z) \geq 0, \tag{2.121}$$

with equality if and only if X and Y are independent when conditioning on Z.

Proof of Lemma 2.41. Assume $X \to Y \to Z$, i.e., $X \perp Z|Y$. By Theorem 2.34, we have

$$I(X;Z|Y) = 0. \tag{2.154}$$

Then

$$\begin{aligned} I(X;Z) &= I(X;Y,Z) - I(X;Y|Z) && (2.155)\\ &\leq I(X;Y,Z) && (2.156)\\ &= I(X;Y) + I(X;Z|Y) && (2.157)\\ &= I(X;Y). && (2.158) \end{aligned}$$

In (2.155) and (2.157), we have used the chain rule for mutual information. The inequality in (2.156) follows because $I(X;Y|Z)$ is always nonnegative, and (2.158) follows from (2.154). This proves (2.149).

Since $X \to Y \to Z$ is equivalent to $Z \to Y \to X$, we also have proved (2.150). This completes the proof of the lemma. □

Data Processing Theorem

Proposition 2.10 (Markov subchains). *Let $\mathcal{N}_n = \{1, 2, \cdots, n\}$ and let $X_1 \to X_2 \to \cdots \to X_n$ form a Markov chain. For any subset α of $\mathcal{N}_n$, denote $(X_i, i \in \alpha)$ by X_α. Then for any disjoint subsets $\alpha_1, \alpha_2, \cdots, \alpha_m$ of $\mathcal{N}_n$ such that*

$$k_1 < k_2 < \cdots < k_m \tag{2.19}$$

for all $k_j \in \alpha_j$, $j = 1, 2, \cdots, m$,

$$X_{\alpha_1} \to X_{\alpha_2} \to \cdots \to X_{\alpha_m} \tag{2.20}$$

forms a Markov chain. That is, a subchain of $X_1 \to X_2 \to \cdots \to X_n$ is also a Markov chain.

Theorem 2.42 (Data Processing Theorem). *If $U \to X \to Y \to V$ forms a Markov chain, then*

$$I(U;V) \le I(X;Y). \tag{2.159}$$

Proof. Assume $U \to X \to Y \to V$. Then by Proposition 2.10, we have $U \to X \to Y$ and $U \to Y \to V$. From the first Markov chain and Lemma 2.41, we have

$$I(U;Y) \le I(X;Y). \tag{2.160}$$

From the second Markov chain and Lemma 2.41, we have

$$I(U;V) \le I(U;Y). \tag{2.161}$$

Combining (2.160) and (2.161), we obtain (2.159), proving the theorem. □

Entropy's upper bound

Theorem 2.43 For any random variable X,

$$H(X) \leq \log |\mathcal{X}|,$$

where $|\mathcal{X}|$ denotes the size of the alphabet $\mathcal{X}$. This upper bound is tight if and only if X is distributed uniformly on $\mathcal{X}$.

Corollary 2.44 The entropy of a random variable may take any nonnegative real value.

Remark:

- The entropy of a random variable is finite as long as it has a finite alphabet.
- However, if a random variable has a countable alphabet, its entropy may or may not be finite.

Proof of the Inequality

Theorem 2.43. *For any random variable X,*

$$H(X) \le \log |\mathcal{X}|, \tag{2.162}$$

where $|\mathcal{X}|$ denotes the size of the alphabet $\mathcal{X}$. This upper bound is tight if and only if X is distributed uniformly on $\mathcal{X}$.

Proof. Let u be the uniform distribution on $\mathcal{X}$, i.e., $u(x) = |\mathcal{X}|^{-1}$ for all $x \in \mathcal{X}$. Then

$$\begin{aligned}
&\log |\mathcal{X}| - H(X) \\
&= -\sum_{x \in \mathcal{S}_X} p(x) \log |\mathcal{X}|^{-1} + \sum_{x \in \mathcal{S}_X} p(x) \log p(x) && (2.163) \\
&= -\sum_{x \in \mathcal{S}_X} p(x) \log u(x) + \sum_{x \in \mathcal{S}_X} p(x) \log p(x) && (2.164) \\
&= \sum_{x \in \mathcal{S}_X} p(x) \log \frac{p(x)}{u(x)} && (2.165) \\
&= D(p\|u) && (2.166) \\
&\ge 0, && (2.167)
\end{aligned}$$

proving (2.162). This upper bound is tight if and if only $D(p\|u) = 0$, which from Theorem 2.31 is equivalent to $p(x) = u(x)$ for all $x \in \mathcal{X}$, completing the proof. □

Corollary 2.44. *The entropy of a random variable may take any nonnegative real value.*

Proof of Corollary 2.44

$$l(a) = \begin{cases} a \log a & \text{if } a > 0 \\ 0 & \text{if } a = 0, \end{cases} \tag{2.73}$$

$$H(p) = -\sum_{x \in \mathcal{X}} l(p(x)), \tag{2.74}$$

Proof. Consider a random variable X defined on a fixed finite alphabet $\mathcal{X}$. We see from the last theorem that $H(X) = \log |\mathcal{X}|$ is achieved when X is distributed uniformly on $\mathcal{X}$. On the other hand, $H(X) = 0$ is achieved when X is deterministic. For $0 \le a \le |\mathcal{X}|^{-1}$, let

$$g(a) = H\left(\{1 - (|\mathcal{X}| - 1)a, a, \cdots, a\}\right) \tag{2.168}$$

$$= -l(1 - (|\mathcal{X}| - 1)a) - (|\mathcal{X}| - 1)l(a), \tag{2.169}$$

where $l(\cdot)$ is defined in (2.73). Note that $g(a)$ is continuous in a, with $g(0) = 0$ and $g(|\mathcal{X}|^{-1}) = \log |\mathcal{X}|$. For any value $0 < b < \log |\mathcal{X}|$, by the intermediate value theorem of continuous functions, there exists a distribution for X such that $H(X) = b$. Then we see that $H(X)$ can take any positive value by letting $|\mathcal{X}|$ be sufficiently large. This accomplishes the proof. □

Example 1 (when the alphabet $\mathcal{X}$ is countable)

Example 2.45. Let X be a random variable such that

$$\Pr\{X = i\} = 2^{-i}, \tag{2.171}$$

$i = 1, 2, \cdots$. Then

$$H_2(X) = \sum_{i=1}^{\infty} i 2^{-i} = 2, \tag{2.172}$$

which is finite.

Example 2 (when the alphabet $\mathcal{X}$ is countable)

Example 2.46. Let Y be a random variable which takes values in the subset of pairs of integers

$$\left\{(i,j) : 1 \leq i < \infty \text{ and } 1 \leq j \leq \frac{2^{2^i}}{2^i}\right\} \tag{2.173}$$

such that

$$\Pr\{Y = (i,j)\} = 2^{-2^i} \tag{2.174}$$

for all i and j. First, we check that

$$\sum_{i=1}^{\infty} \sum_{j=1}^{2^{2^i}/2^i} \Pr\{Y = (i,j)\} = \sum_{i=1}^{\infty} 2^{-2^i} \left(\frac{2^{2^i}}{2^i}\right) = 1. \tag{2.175}$$

Then

$$H_2(Y) = -\sum_{i=1}^{\infty} \sum_{j=1}^{2^{2^i}/2^i} 2^{-2^i} \log_2 2^{-2^i} = \sum_{i=1}^{\infty} 1, \tag{2.176}$$

which does not converge.

Fano's Inequality

Let X be a random variable and $\hat{X}$ be an estimate on X which takes value in the same alphabet $\mathcal{X}$. Let the probability of error P_e be

$$P_e = \Pr\{X \neq \hat{X}\}. \tag{2.177}$$

If $P_e = 0$, i.e., $X = \hat{X}$ with probability 1, then $H(X|\hat{X}) = 0$ by Proposition 2.36. Intuitively, if P_e is small, i.e., $X = \hat{X}$ with probability close to 1, then $H(X|\hat{X})$ should be close to 0. Fano's inequality makes this intuition precise.

Theorem 2.47 (Fano's Inequality) Let X and $\hat{X}$ be random variables taking values in the same alphabet $\mathcal{X}$. Then

$$H(X|\hat{X}) \leq h_b(P_e) + P_e \log(|\mathcal{X}| - 1),$$

where h_b is the binary entropy function.

Proof of Fano's Inequality

Theorem 2.47 (Fano's Inequality) Let X and $\hat{X}$ be random variables taking values in the same alphabet $\mathcal{X}$. Then

$$H(X|\hat{X}) \leq h_b(P_e) + P_e \log(|\mathcal{X}| - 1),$$

where h_b is the binary entropy function.

Proof. Define a random variable

$$Y = \begin{cases} 0 \text{ if } X = \hat{X} \\ 1 \text{ if } X \neq \hat{X}. \end{cases} \tag{2.179}$$

The random variable Y is an indicator of the error event $\{X \neq \hat{X}\}$, with $\Pr\{Y = 1\} = P_e$ and $H(Y) = h_b(P_e)$. Since Y is a function X and $\hat{X}$,

$$H(Y|X, \hat{X}) = 0. \tag{2.180}$$

Then

$$\begin{aligned}
&H(X|\hat{X}) \\
&= H(X|\hat{X}) + H(Y|X, \hat{X}) && (2.181)\\
&= H(X, Y|\hat{X}) && (2.182)\\
&= H(Y|\hat{X}) + H(X|\hat{X}, Y) && (2.183)\\
&\leq H(Y) + H(X|\hat{X}, Y) && (2.184)\\
&= H(Y) + \sum_{\hat{x} \in \mathcal{X}} \Big[\Pr\{\hat{X} = \hat{x}, Y = 0\} H(X|\hat{X} = \hat{x}, Y = 0) \\
&\quad + \Pr\{\hat{X} = \hat{x}, Y = 1\} H(X|\hat{X} = \hat{x}, Y = 1) \Big]. && (2.185)
\end{aligned}$$

Theorem 2.47 (Fano's Inequality) Let X and $\hat{X}$ be random variables taking values in the same alphabet $\mathcal{X}$. Then

$$H(X|\hat{X}) \leq h_b(P_e) + P_e \log(|\mathcal{X}| - 1),$$

where h_b is the binary entropy function.

Proof of Fano's Inequality

In the above, (2.181) follows from (2.180), (2.184) follows because conditioning does not increase entropy, and (2.185) follows from an application of (2.43). Now X must take the value $\hat{x}$ if $\hat{X} = \hat{x}$ and $Y = 0$. In other words, X is conditionally deterministic given $\hat{X} = \hat{x}$ and $Y = 0$. Therefore, by Proposition 2.35,

$$H(X|\hat{X} = \hat{x}, Y = 0) = 0. \tag{2.186}$$

If $\hat{X} = \hat{x}$ and $Y = 1$, then X must take a value in the set $\{x \in \mathcal{X} : x \neq \hat{x}\}$ which contains $|\mathcal{X}| - 1$ elements. By Theorem 2.43, we have

$$H(X|\hat{X} = \hat{x}, Y = 1) \leq \log(|\mathcal{X}| - 1), \tag{2.187}$$

where this upper bound does not depend on $\hat{x}$. Hence,

$$\begin{aligned} & H(X|\hat{X}) \\ & \leq h_b(P_e) + \left(\sum_{\hat{x} \in \mathcal{X}} \Pr\{\hat{X} = \hat{x}, Y = 1\}\right) \log(|\mathcal{X}| - 1) && (2.188) \\ & = h_b(P_e) + \Pr\{Y = 1\} \log(|\mathcal{X}| - 1) && (2.189) \\ & = h_b(P_e) + P_e \log(|\mathcal{X}| - 1), && (2.190) \end{aligned}$$

which completes the proof. □

Fano's Inequality

Theorem 2.47 (Fano's Inequality) Let X and $\hat{X}$ be random variables taking values in the same alphabet $\mathcal{X}$. Then

$$H(X|\hat{X}) \leq h_b(P_e) + P_e \log(|\mathcal{X}| - 1),$$

where h_b is the binary entropy function.

Corollary 2.48 $H(X|\hat{X}) < 1 + P_e \log |\mathcal{X}|$.

Interpretation

- For finite alphabet, if $P_e \to 0$, then $H(X|\hat{X}) \to 0$.
- This may NOT hold for countably infinite alphabet (see Example 2.49).

Thank you!